# SIMATS ENGINEERING

| Name | Register Number | Course Code | Course |
|---|---|---|---|
| S Susmitha | 192571068 | ITA0510 | Computer Vision |

**19. Perform Edge detection using Sobel Matrix along XY axis.**

**AIM:**
Edge detection using Sobel Matrix along XY axis.

**PROGRAM:**

```
import cv2

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png', 0)

if img is None:
    print("Error: Image not found!")
else:
    # Apply Sobel filter in X and Y directions
    sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
    sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Combine the Sobel X and Sobel Y images
    edges = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

    # Convert to 8-bit image for display and saving
    edges = cv2.convertScaleAbs(edges)

    # Save the output image
    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Edge_detection.jpg',
        edges
    )

    # Display the images
    cv2.imshow("Original Image", img)
    cv2.imshow("Edge Detection", edges)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

**INPUT:**

**OUTPUT:**

**RESULT:**

Thus the program is executed successfully.